

MASTER'S STUDY PROPOSAL 碩士論文計畫書

# The Invisible Ledger

Quantifying the Invisible Wedge in Indonesia's Platform Economy

隱形帳簿：量化印尼平台經濟中的隱形缺口

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Yuan Ze University · Master of Science in Finance · 1 October 2026

## THE PUZZLE

# Accounts said growth. Transactions said contraction.

帳面成長，交易萎縮

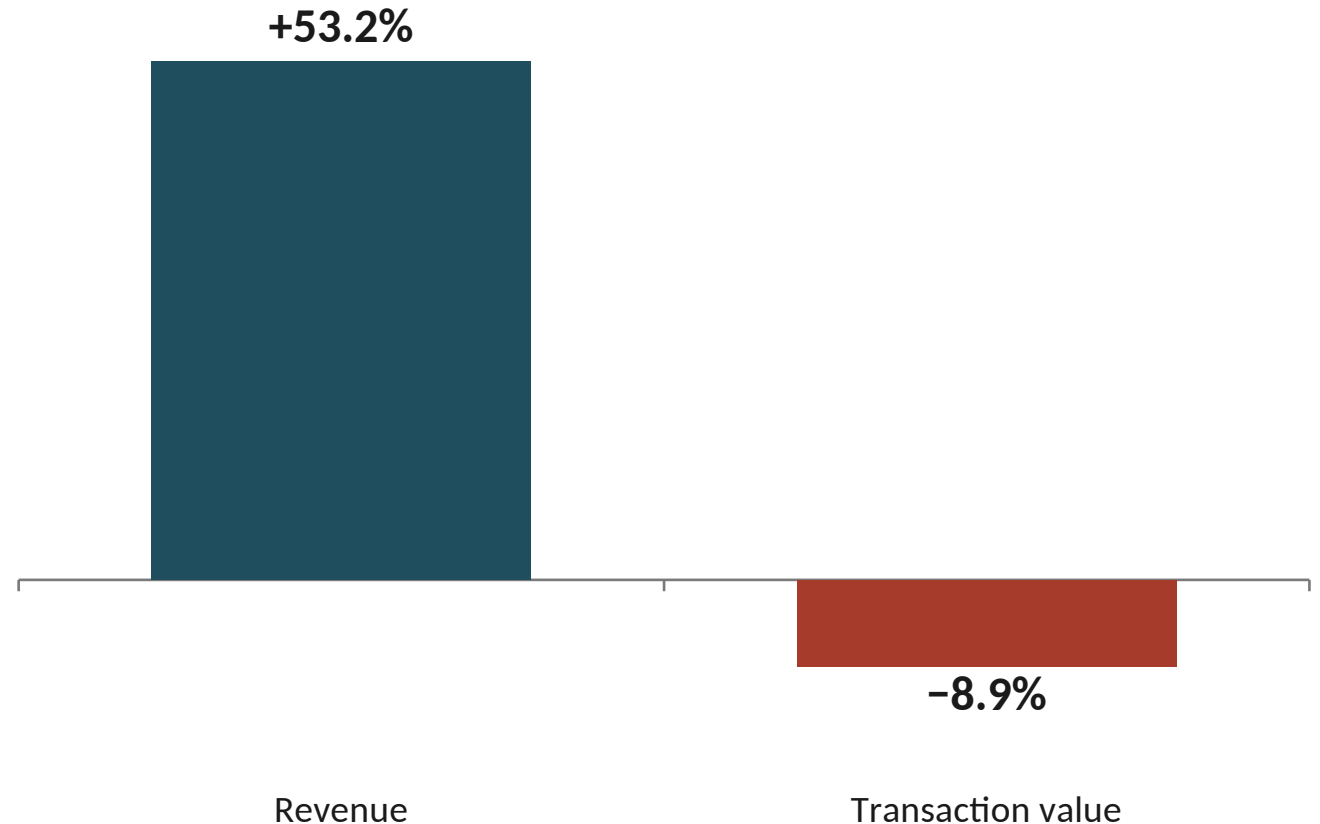
**+53.2%**

Revenue — what the accounts record

**-8.9%**

Transaction value — what the platform carried

**Tokopedia, 2022 → 2023. Same company, same year, both correct.**



# Why revenue rose while transactions fell

營收為何在交易下滑時仍成長

60.6%

Lower customer incentives

39.4%

Higher gross revenue

Share of the 2022–2023 increase in Tokopedia net revenue (+53.2%)

## What are customer incentives?

The partner incentives and end-user promotions — discounts, vouchers, shipping subsidies — that GoTo deducts from gross revenue to arrive at net revenue.

*The platform earned more mainly by spending less to attract transactions — not by carrying more of them.*

# Research question

研究問題與目標

*How much economic activity remains invisible when digital platforms are measured through their reported revenue — and when does that choice change the conclusion about growth?*

## 1 Measure

The transaction–revenue relationship across Indonesian platforms, 2020–2025, using W, E and D.

## 2 Reconcile

Large divergences to disclosed components: incentives, monetization, revenue recognition, scope.

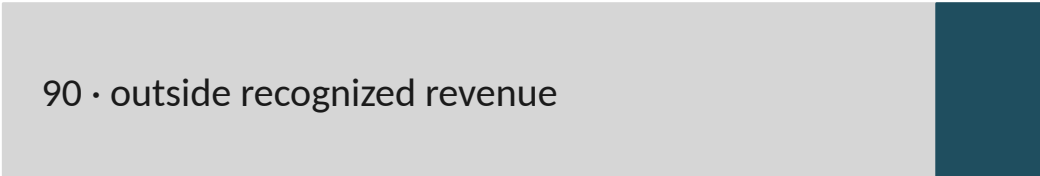
## 3 Compare

Platform evidence against national e-commerce (BPS) and payment (Bank Indonesia) indicators.

# Three measures — and D is the test

三個衡量指標：D 是核心檢定

A platform carries 100 and keeps 10



90 · outside recognized revenue

$V = 100$  (transaction value)    $R = 10$  (revenue)

$W = 90$     $E = 90 / 10 = 9$

*A large but stable gap still preserves the growth story. A moving gap does not.*

$$W = V - R$$

How large the gap is

$$E = (V - R) / R = 1/t - 1$$

The gap relative to revenue — a transformation of the take rate, not the contribution

$$D = g(R) - g(V)$$

Whether the relationship holds over time — the central test. A sign flip is the strongest failure.

# What I can observe — and how directly

樣本與證據層級

## Issuer-reported

Main sample

Tokopedia 2022–23 · Blibli 2021–25 · Bukalapak 2020–23

**11 platform-years**  
**8 comparisons**

## Constructed

2023 scale only

Grab 2021–23 · Shopee 2022–24

Ratios fixed by  
company-wide figures

## Comparison indicators

Never pooled

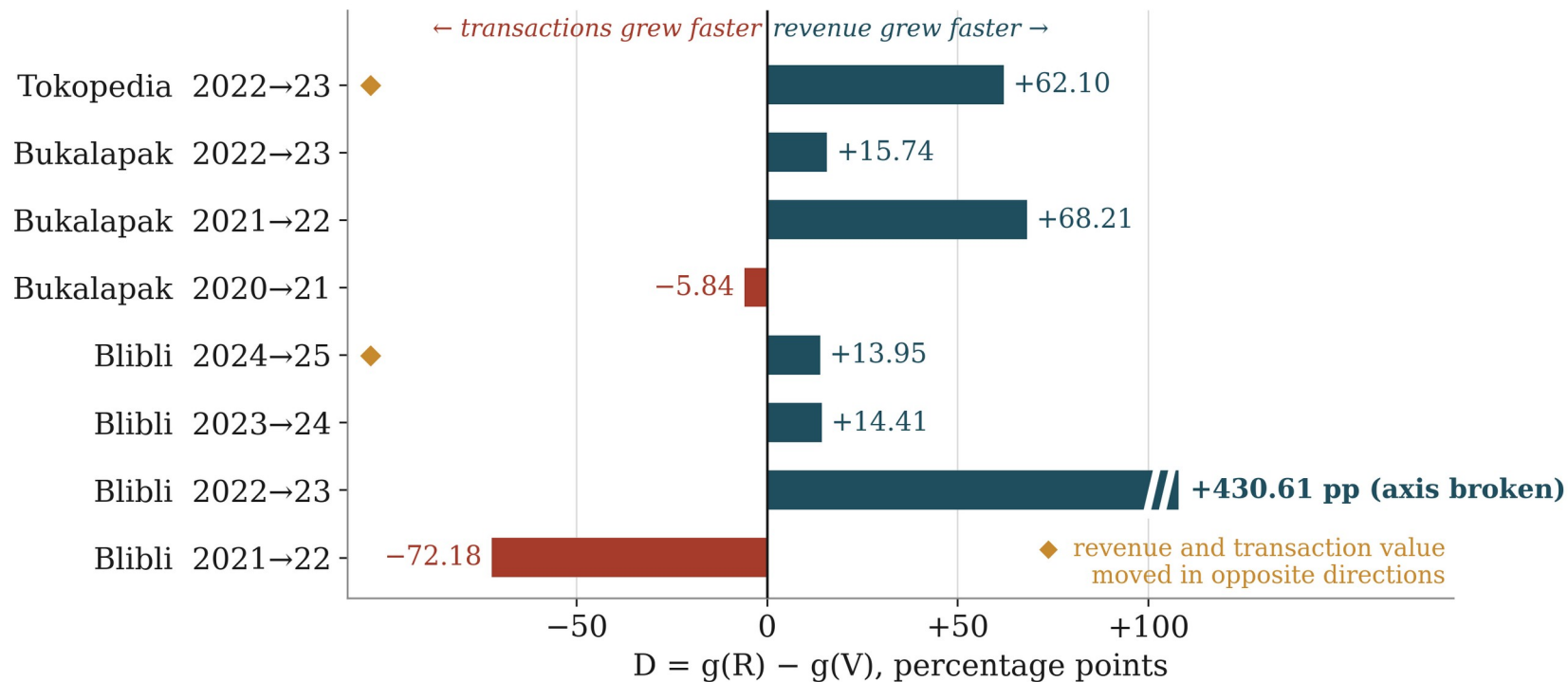
BPS national e-commerce · Bank Indonesia payments · 8 listed  
platforms outside Indonesia

**Context and  
external benchmark**

## THE EVIDENCE

# Revenue outgrew transactions in 6 of 8 comparisons

八次比較中有六次營收成長快於交易額



6 / 8

revenue grew faster

2

full reversals (◆)

38.92 pp

median |D|

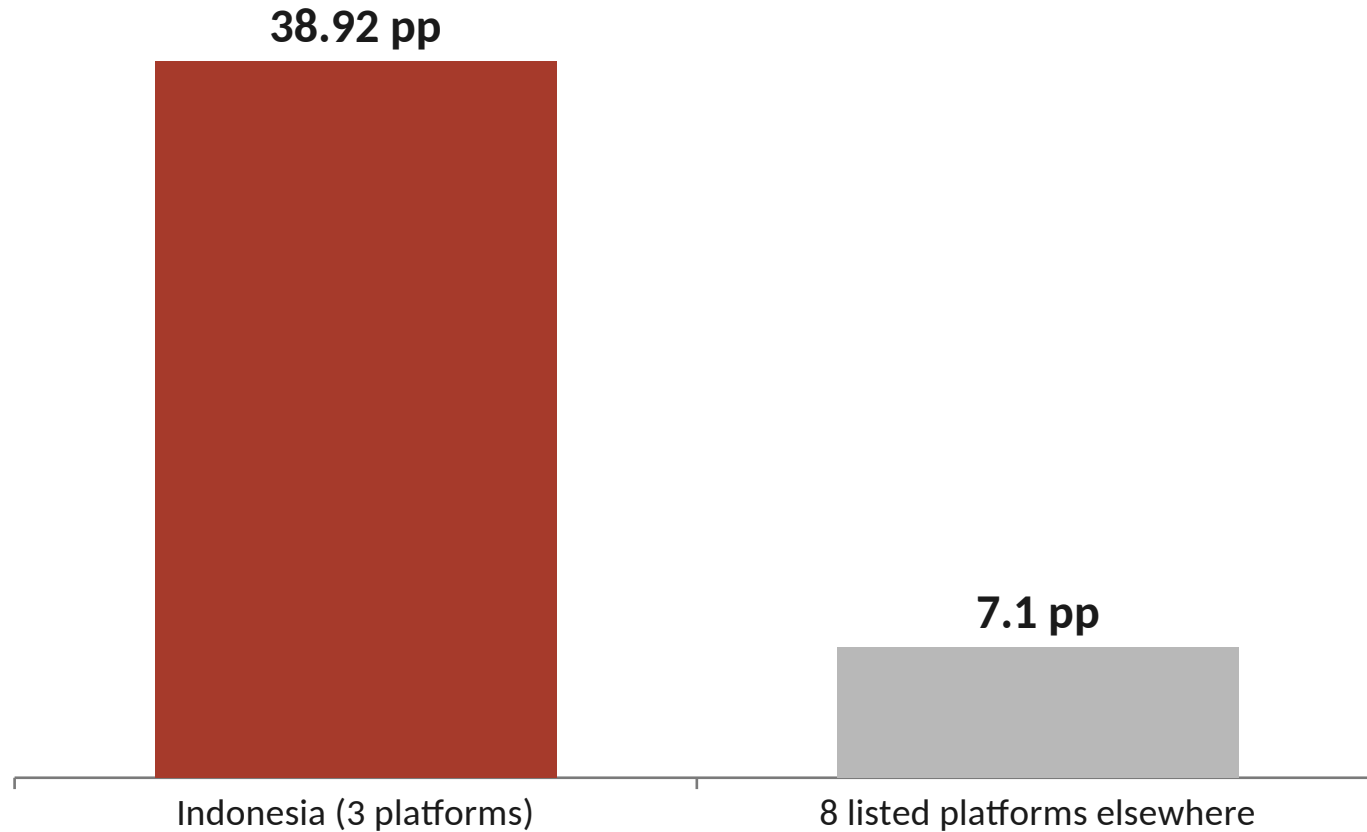
**Without Blibli 2022-23: 5 of 7, median 15.74 pp. Every Ecosystem Ratio ends below where it began.**

## THE EVIDENCE

# The direction is common. The magnitude is not.

方向普遍，幅度不同

Median absolute divergence  $|g(R) - g(V)|$



### Direction: common

Revenue grew faster in 6 of 8 comparisons in Indonesia, and 22 of 29 abroad.

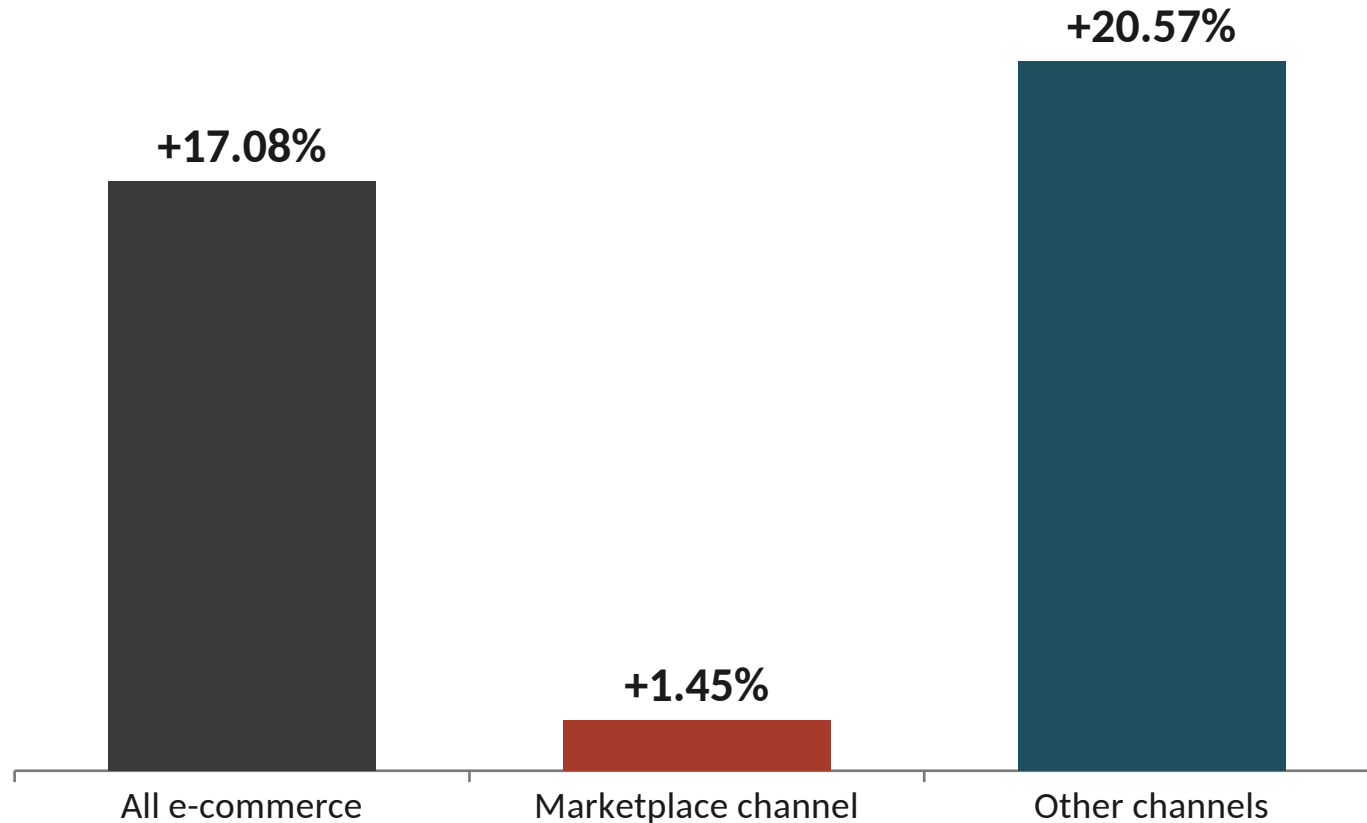
### Magnitude: not

The typical gap is about five times larger in Indonesia: 38.92 vs 7.1 points.

# Nationally, growth happened outside the marketplace channel

全國電商成長主要發生在市集通路之外

Growth in value, 2023 → 2024



**98.46%**

of the 2023–24 increase in e-commerce value fell outside BPS's marketplace channel

BPS classifies surveyed businesses' online sales by channel: marketplaces, social media, instant messaging, own website.

## THE EVIDENCE

# Growth came from new sellers, not bigger sellers

成長來自新進賣家，而非既有賣家擴張

**+17.08%**

e-commerce value

=

**+15.31%**

number of businesses

×

**+1.54%**

sales per business

**90.29%**

of the increase came from more businesses selling online

70.95% under BPS's alternative 2023 business count — entry dominates either way.

Marketplace sellers keep better records

**28.63%**

of marketplace users

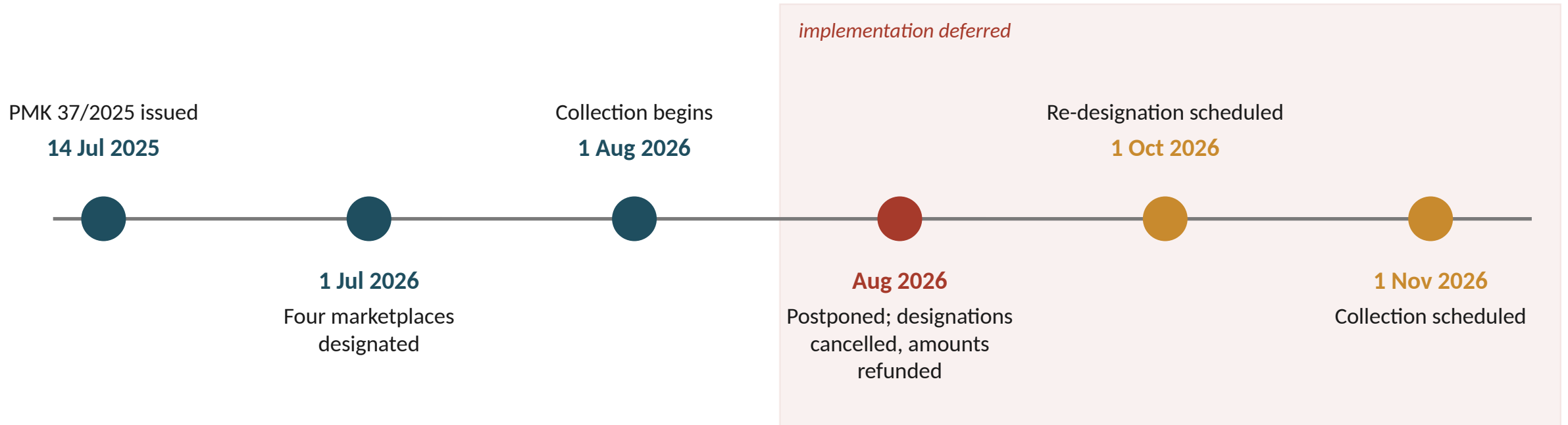
**12.25%**

of non-users

*keep complete financial statements — an association, not a cause*

# Marketplace records are becoming tax infrastructure

市集紀錄正成為稅務行政基礎



Tax administration is being routed through the marketplace record — while 98% of measured e-commerce growth sat outside the marketplace channel. Different categories; worth seeing side by side. Not a forecast.

# Two kinds of revenue growth look identical in the accounts

兩種營收成長在帳上看起來相同

## From more commerce

Transactions grow; revenue follows.

**Can continue**

## From fewer incentives

Revenue grows while transactions stall or fall — Tokopedia 2022–23.

**Bounded**

*a platform can stop discounting only once*

*Reported revenue shows the same number for both. The finance question: is revenue a reliable signal of a platform's underlying growth?*

Proposed extension (Prof. Kong's suggestion): which indicator, or revenue-growth component, is more informative for stock returns — exploratory, given a small, segment-affected listed sample.

# What the evidence does — and does not — establish

證據能說明與不能說明的範圍

| Establishes                                                         | Does not by itself establish                                      |
|---------------------------------------------------------------------|-------------------------------------------------------------------|
| Revenue and transaction value move apart within platforms (W, E, D) | Participant profit, taxable income, unpaid tax, or missing GDP    |
| Disclosed components reconcile selected movements                   | A causal effect of incentives, monetization or accounting choices |
| National channels and platform records grow at very different rates | Which indicator, if any, contains measurement error               |
| PMK 37/2025 draws on designated operators' records                  | Its coverage of the digital economy, or any compliance effect     |

*Every record is correct for what it measures. The error appears only when one is used in place of another.*

NEXT

# Plan to the final defence

邁向學位考試的工作規劃

1

## Proposal stage

Incorporate committee feedback.  
Finalize source definitions and figures.

2

## Oct – Nov 2026

Reconcile BPS and platform definitions. Split movements into volume and monetization effects. Alternative sample rules; robustness.

3

## Before the final defence

Complete the manuscript. Decide and, if feasible, run the stock-return extension.

The data and framework are assembled; what remains is reconciliation, robustness and the manuscript.

*The same company can look like it is growing or shrinking depending on which of its own numbers you read.*

This study shows when that happens, why, and that it also appears in the national statistics — as Indonesia begins building tax collection on the more visible record.

**Thank you 謝謝**

# Backup

Scale • construction • ratio paths • BPS levels • payments • hypotheses • Tokopedia bridge

## 2023 scale: a large level gap

| Case (US\$ bn)         | V             | R            | W             | E              | Evidence                                |
|------------------------|---------------|--------------|---------------|----------------|-----------------------------------------|
| Grab Indonesia         | 5.381         | 0.605        | 4.776         | 7.895×         | Transaction value constructed           |
| Tokopedia              | 16.331        | 0.405        | 15.926        | 39.296×        | Issuer-reported pair                    |
| Shopee Indonesia       | 21.520        | 2.152        | 19.368        | 9.000×         | Transaction value and revenue estimated |
| <b>Three platforms</b> | <b>43.233</b> | <b>3.162</b> | <b>40.070</b> | <b>12.671×</b> | Sum of three cases, not national total  |

≈ 2.9% of Indonesia's 2023 nominal GDP (US\$1.371 trillion) — magnitude only; the wedge is transaction value, not value added.  
 Shopee rate varied 9–11% → W between US\$39.86B and US\$40.29B. Excluding any one platform leaves at least US\$20.70B.

# Why Shopee is 9.000× and Grab 7.9×

## Shopee Indonesia

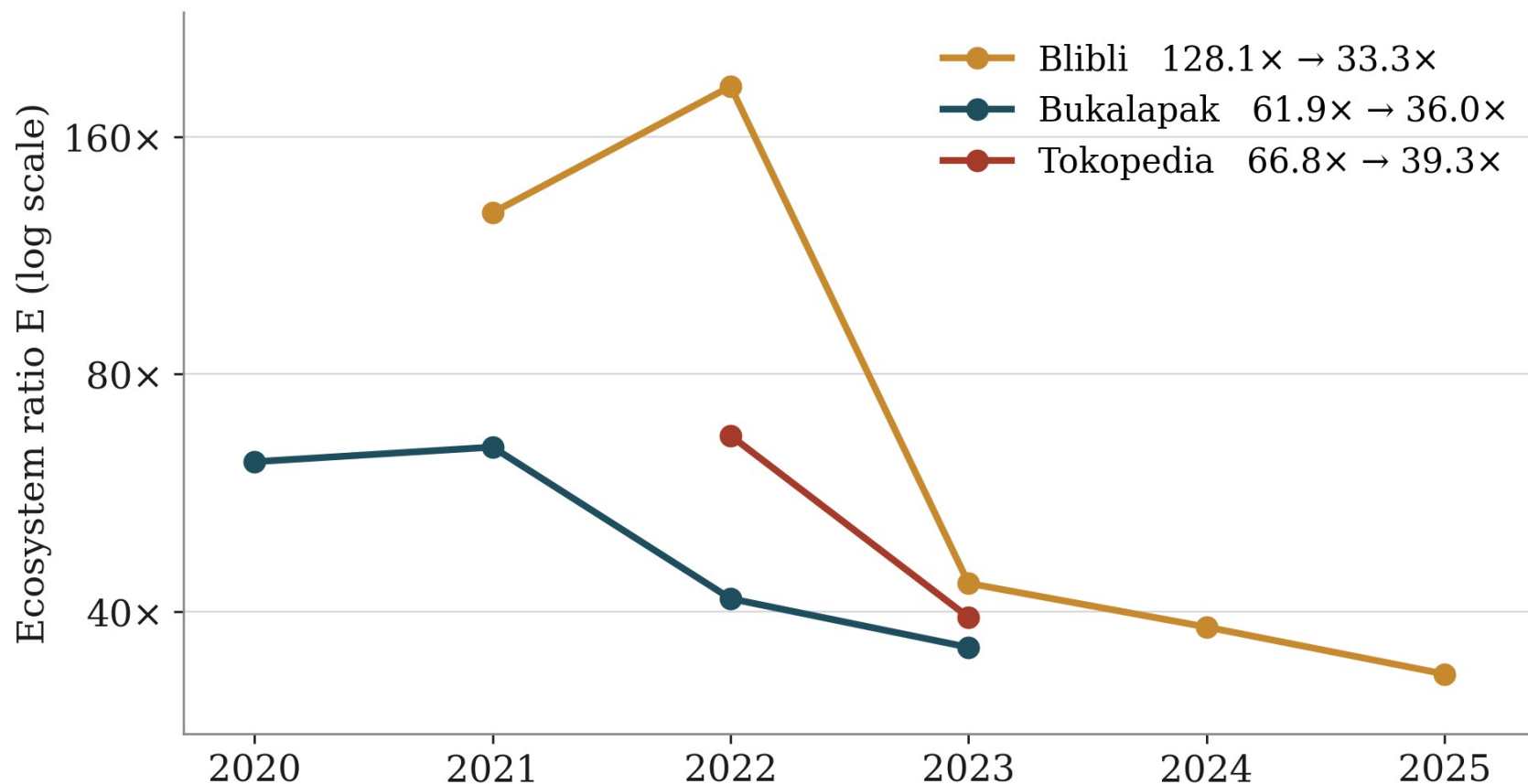
Transaction value: third-party estimate (Momentum Works). Revenue: that estimate × Sea's company-wide monetization rate (assumed 10%). So  $E = 1/0.10 - 1 = 9.000$  by construction.

## Grab Indonesia

Indonesian revenue is disclosed; transaction value = Indonesian revenue × Group GTV / Group revenue. So Grab's Indonesian E equals its Group E (7.9×).

**Both contribute 2023 levels only — no independent evidence on the ratio or its movement. The time-series result uses issuer-reported pairs alone.**

## Ecosystem ratio over time (issuer-reported, log scale)



Years without a matched pair omitted (Tokopedia 2021; Bukalapak 2024: nine months of transactions vs twelve of revenue).

## BPS levels are not like-for-like with platform figures

**Rp200.68T**

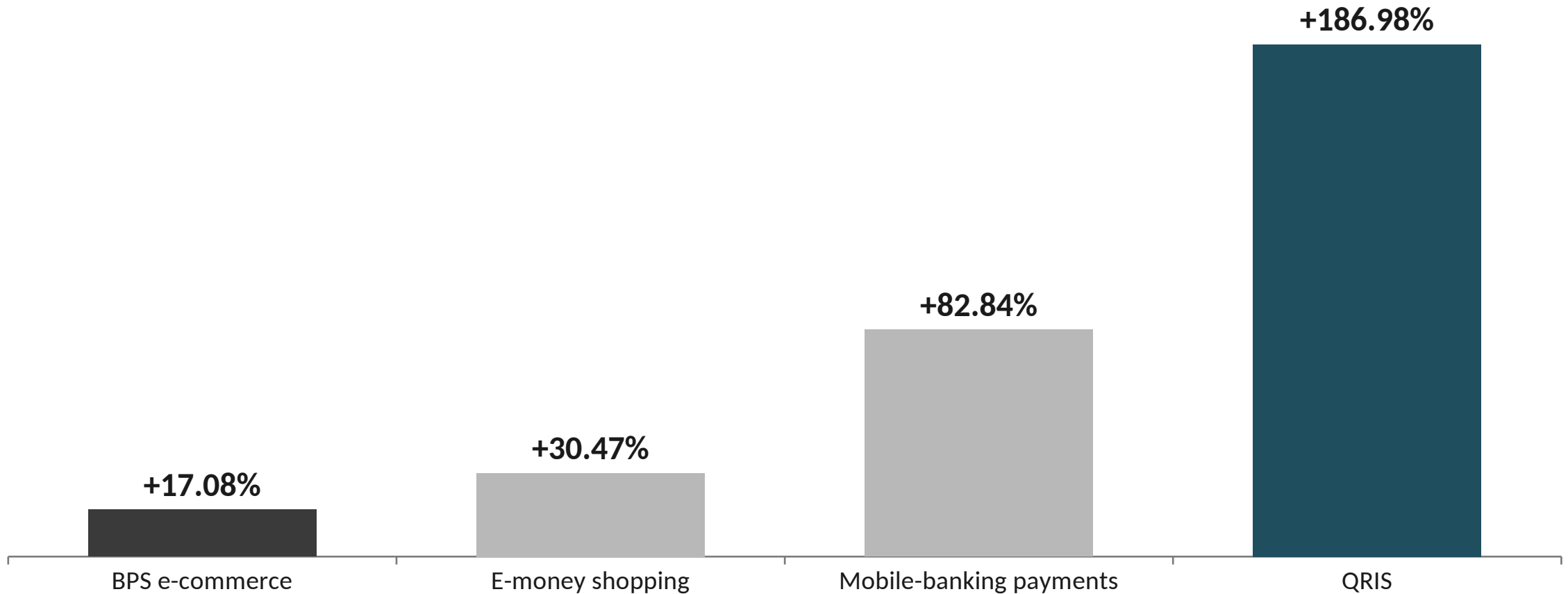
BPS 2023 marketplace component ≈ US\$13.2B (18.2% of e-commerce, exclusive channel split)

**US\$37.9B**

Tokopedia reported GTV + external Shopee GMV estimate, 2023

Different definitions and channel attribution — the study reconciles them before attributing any gap to coverage.  
BPS multiple-response measure (social media 44.04%, marketplaces 32.74% of value) is not comparable with the 18.2% exclusive split.

# Payment measures grew far faster than e-commerce value



# Hypotheses and principal tests

| Hypothesis                                                                   | Principal test                                                                       |
|------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| H1 Within platforms, transaction value and revenue do not grow in proportion | D and E for issuer-reported pairs: level, stability, sign reversals                  |
| H2 National e-commerce growth comes mainly from more businesses              | BPS value split into business count and sales per business                           |
| H3 Marketplace participation is associated with stronger recordkeeping       | BPS comparison of marketplace and non-marketplace sellers                            |
| H4 Conclusions about growth change with the indicator used (central claim)   | Revenue vs transaction value within platforms; marketplace channel vs all e-commerce |

## Tokopedia revenue bridge, 2022–2023 (Rp trillion)

|                                | FY2022      | FY2023      | Change                |
|--------------------------------|-------------|-------------|-----------------------|
| Third-party gross revenue      | 8.14        | 8.99        | +0.85 (39.4%)         |
| Customer incentives (deducted) | -4.11       | -2.81       | +1.30 (60.6%)         |
| <b>Third-party net revenue</b> | <b>4.03</b> | <b>6.18</b> | <b>+2.14 (+53.2%)</b> |

Net revenue change = gross revenue change + reduction in incentives. Incentives fell by about a third while transaction value fell 8.9%.